Row Across Time: A Tangible Ba He Tong VR Game for Embodied Cultural Play

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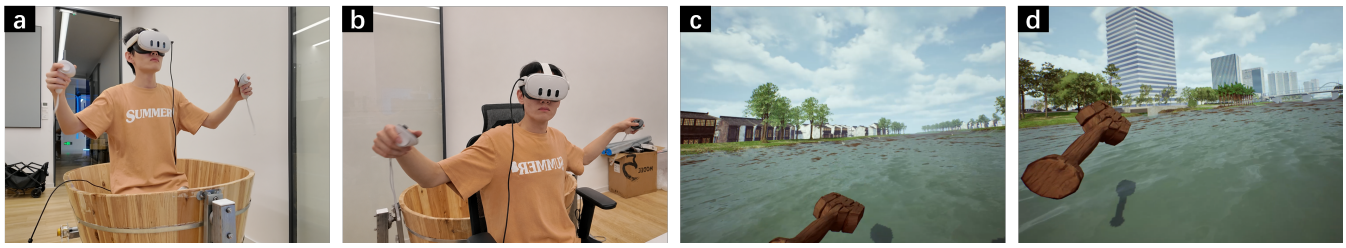


Figure 1: *Row Across Time*: (a) the tangible *Ba He Tong* condition, where the player sits inside the physical vessel; (b) the virtual-only condition, where the same VR game is played from a regular chair; (c) a first-person rowing view in Old Jiaomen River; and (d) a first-person rowing view in Present Jiaomen River.

Abstract

Local cultural practices are difficult to communicate when they are presented only as images, models, or explanatory text. *Ba He Tong* is a local Pearl River Delta water activity in which people paddle a wooden grain-tub-like vessel. To make this practice playable, we introduce *Row Across Time*, a short virtual reality (VR) rowing game set on Jiaomen River in Nansha, Guangzhou, in Southern China. Players sit inside a one-to-one tangible replica of the *Ba He Tong* vessel, wear a VR headset, and row from an old water-town interpretation of Jiaomen River to a present-day urban waterfront. The game uses one repeated seated rowing action as locomotion, interaction, and cultural participation. We describe the game design, tangible interface, implementation, and formative evaluation. In a within-subjects study with 12 participants, the tangible setup significantly improved presence, embodied rowing, cultural exploration, artifact experience, and enjoyment compared with a virtual-only seated setup of the same game. Completion time and discomfort did not differ significantly. These results suggest that a cultural artifact can function not only as visual content in VR, but also as the physical interface through which players interpret a place-based cultural journey.

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CCS Concepts

• **Applied computing** → **Computer games**; • **Human-centered computing** → **Virtual reality**; *Haptic devices.*

Keywords

virtual reality, tangible interaction, cultural heritage, embodied interaction, rowing games

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1 Introduction

Cultural heritage is often encountered through museums, photographs, video records, reconstructed environments, or short textual explanations. These forms can show what an artifact looks like, but they do not always help people understand what it feels like to use the artifact or to perform the practice around it. In digital cultural heritage, augmented, virtual, and mixed reality have been widely explored because they can place visitors inside otherwise inaccessible spaces and historical interpretations [2, 3, 13]. However, when interaction is limited to looking around, moving through a scene, or collecting information, the player's body can remain separate from the cultural practice being represented [18].

The above-mentioned aspect is important because *Ba He Tong* and similar practices are not merely objects to be viewed. It is a water activity associated with paddling a wooden grain-tub-like vessel. The cultural meaning of the activity comes from a relationship between body, vessel, water, and historic place. We therefore asked how a VR game could make *Ba He Tong* understandable through bodily participation rather than exposition alone.

Row Across Time is our answer to this question. It is a short VR rowing game set on Jiaomen River in Nansha, Guangzhou, a main city in Southern China, located next to Hong Kong. The player rows through two stylized versions of the river: Old Jiaomen River, which presents a quieter water-town environment, and Present Jiaomen River, which presents an urban waterfront. The player uses the same rowing motion throughout the experience. This continuous action is intentionally simple. It gives players time to attend to the river, the vessel, and the change in surroundings.

The central design element is a one-to-one tangible *Ba He Tong* replica. In the tangible version, players physically sit inside the wooden vessel while seeing and controlling its virtual counterpart. We use *virtual-only* to refer to a seated VR version without the physical *Ba He Tong* installation: players sit on a regular chair, still wear the VR headset, and use the same controller-based rowing input. This comparison allowed us to examine whether physically occupying the cultural artifact changes how the same VR game is experienced.

We present the design and evaluation of *Row Across Time*. The contribution is threefold: (1) a playable VR rowing game that communicates Jiaomen River through an old-to-present journey; (2) a tangible *Ba He Tong* interface that connects seated posture, rowing movement, and cultural meaning; and (3) formative user-study evidence showing that the tangible interface strengthens the intended experiences of presence, embodied rowing, cultural exploration, artifact experience, and enjoyment.

2 Background and Related Work

2.1 Interactive Cultural Heritage

Virtual heritage and serious games have been used to support cultural learning, historical interpretation, and engagement with places and artifacts [2, 3, 12, 13]. Compared with static presentation, interactive media can support learning by doing and exploration through play [14]. Yet interactivity alone does not guarantee cultural understanding. A player may move through a reconstructed environment without performing an action that is meaningfully connected to the represented practice.

This project, therefore, focuses on the relation between action and cultural interpretation. Jiaomen River is not presented as a complete historical reconstruction. Instead, it is organized as a short playable journey between two interpretive scenes. The goal is to help a general audience feel a bodily relationship to a local water practice, using rowing as the repeated action that connects the player to the vessel and the river.

2.2 Tangible and Embodied Interaction in VR

Tangible interaction emphasizes the connection between digital information, physical form, and situated action [5, 7, 9, 11, 20]. In VR, physical props and passive haptics can increase plausibility by aligning what users see with what they touch [1, 8, 16, 19]. Presence and embodiment research further shows that users' sense of being there and acting through a body can shape how they experience virtual environments [10, 15, 17].

Our design builds on these ideas but uses the physical object in a specific way. The tangible *Ba He Tong* is not only a prop that improves realism. It is the same cultural object that the game asks players to understand. By placing the player inside the vessel before the VR scene begins, the installation frames the whole experience as a physical encounter with a local water practice.

3 Game Design

3.1 Game Concept and Target Experience

Row Across Time is a short VR rowing experience set on Jiaomen River, the main river that runs through Nansha, a suburb located in the southern part of Guangzhou. The prototype was developed in Unreal Engine 5.5 and runs on a PC-linked Meta Quest 3 setup. It is designed for short play sessions in classroom demonstrations, campus showcases, local cultural exhibitions, and public events. The target player is a general audience member who can safely use a VR headset and perform light seated upper-body movement. A complete playthrough lasts approximately 2–5 minutes.

The player takes the role of a rower traveling along Jiaomen River across time. The target experience is calm and reflective rather than competitive. Players row forward, steer the vessel, observe the riverbank, and reach the end of each route. The design avoids adding separate tasks that would pull attention away from the bodily relationship between the player, the vessel, and the river.

3.2 Tangible *Ba He Tong* Design

The tangible installation is a one-to-one wooden *Ba He Tong* replica built for the game. Figure 2 shows the relation between traditional wooden tubs used in *Ba He Tong* activities, our physical game

installation, and the virtual vessel. The installation uses a circular tub-like enclosure, wooden material, and visible metal bands. These features are intended to make the player feel situated inside a specific vessel rather than seated on a generic VR chair.

The installation serves two connected purposes. As an interface, it grounds the player's body, constrains posture, and supports rowing. As a cultural artifact, it gives the player a physical point of entry into the experience. The object is visible and touchable before the headset is worn, so the cultural theme begins in the real room rather than only inside the virtual scene.

3.3 Rowing as the Core Interaction

The core loop is simple: look ahead, row with the left and right controllers, steer the vessel, observe the environment, and continue the journey. The two VR controllers function as paddles. Players do not need to press buttons or learn additional input mappings; movement is controlled entirely through paddle-like controller motions. Balanced left and right strokes move the vessel forward, while stronger strokes on one side rotate it.

This button-free interaction reduces the learning cost and task load for first-time players. It also keeps players' attention on the relationship between their arm movement, the vessel, and the river environment rather than on operating a controller interface. The rowing controller, therefore, prioritizes stability and readability over physical simulation. Players should quickly understand how their movement affects the boat, while the repeated motion remains meaningful enough to support the cultural theme. Rowing becomes more than locomotion: it is the player's primary means of engaging in the represented water cultural practice.

3.4 Two-Scene River Journey

The game contains two sequential scenes (Figure 3). Both scenes use the same rowing controls and a clear forward route. They differ in architecture, riverbank objects, atmosphere, and spatial composition. This structure lets players feel a sense of continuity in their own bodily action while the surrounding river changes.

The **Old Jiaomen River** presents a quiet water-town route. Low riverside houses and fields evoke a slower river life. The player rows at a comfortable pace and observes the scene from inside the vessel. An invisible finish line is placed near the end of the route. When the player reaches it, the game automatically transitions to the **Present Jiaomen River** scene after a brief loading interval of about one second. This makes the scene change feel like a continuous river journey rather than a menu-based level selection.

The **Present Jiaomen River** shifts to a contemporary waterfront. The riverbanks contain bridges, greenways, park-like landscape elements, and urban buildings. Since the player continues the same rowing action, the scene contrast comes from the surrounding place rather than from a new game task. The resulting experience is a continuous journey through changing interpretations of the same river.

4 User Study and Evaluation

4.1 Participants and Procedure

We conducted a within-subjects formative study with 12 participants (5 women, 7 men), aged 21–27 years ($M = 23.92$, $SD = 1.83$). Participants reported low to moderate familiarity with VR games ($M = 2.58$, $SD = .79$, 1–7 scale), low familiarity with Jiaomen River or Nansha water-town culture ($M = 1.58$, $SD = .90$), and no prior familiarity with *Ba He Tong* ($M = 1.00$, $SD = .00$). This sample was appropriate for evaluating whether the experience could introduce the artifact and place to players who did not already know them.

Each participant experienced both conditions: tangible *Ba He Tong*, where players sat on the customized wooden boat, and virtual-only *Ba He Tong*, where the same VR game was played with participants sitting on a regular chair. The order was counterbalanced. Before the first condition, participants completed a short background questionnaire and watched an introduction video about Jiaomen River and the *Ba He Tong* activity[1]. After each condition, they completed a 7-point post-condition questionnaire, and their completion time was logged.

4.2 Measures and Analysis

The measurements most directly connected to the design purpose were: presence, embodied rowing, cultural exploration, artifact experience, enjoyment, discomfort, and completion time. Table 1 lists the constructs and representative items. Multi-item construct scores were computed by averaging the relevant items. Internal consistency was good for the reported multi-item constructs (Cronbach's $\alpha = .81 - .93$). We used paired-sample t-tests to compare conditions, applied Holm correction across the reported comparisons, and report Cohen's d_z as a paired effect size. The study is intended as formative game-design evidence rather than a broad psychological claim.

Table 1: Reported measures and representative items. All questionnaire items used 7-point scales.

Measure	Representative item
Presence	"I felt like I was inside the river environment."
Embodied rowing	"My body movement felt meaningfully connected to the movement of the vessel."
Cultural exploration	"This experience felt like cultural exploration rather than only a rowing game."
Artifact experience	"The vessel felt like a culturally meaningful object rather than a generic game vehicle."
Enjoyment	"I enjoyed this experience."
Discomfort	"Please rate your symptoms after this condition - general discomfort."

4.3 Results

The tangible setup improved all five focal experience dimensions (Figure 4; Table 2). Presence increased from $M = 4.33$ to $M = 5.75$, embodied rowing from $M = 4.92$ to $M = 6.22$, cultural exploration from $M = 4.54$ to $M = 5.79$, artifact experience from $M = 4.81$ to $M = 6.00$, and enjoyment from $M = 5.54$ to $M = 6.58$. All five differences remained significant after Holm correction. These results suggest that the physical *Ba He Tong* strengthened the intended experiential and cultural qualities of the same VR game.

[1]The excerpt corresponded to 0:00–1:28 of the publicly available YouTube video *Solo Dragon Boating? So Exciting!*, uploaded by Amazing Zhongshan. URL: https://www.youtube.com/watch?v=To6FVKv3VNQ. Accessed May 2026.



Figure 2: Tangible and virtual *Ba He Tong* design: (a) traditional wooden tubs used in *Ba He Tong* activities; (b) our one-to-one tangible replica for the VR game; and (c) the virtual *Ba He Tong* model used in the game.

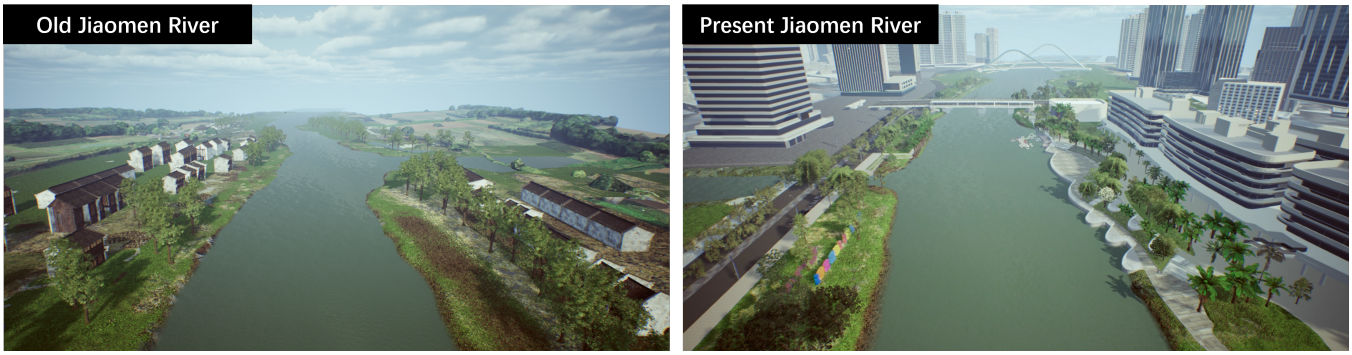


Figure 3: The two scene routes contrast an old water-town interpretation of Jiaomen River with a present-day urban waterfront. The same rowing action carries the player through both scenes.

Table 2: Within-subjects results. p_H indicates Holm-corrected p-values across the reported comparisons.

Measure	Virtual-only M(SD)	Tangible M(SD)	Diff.	t(11)	p_H	d_z
Presence	4.33 (.28)	5.75 (.41)	1.42	9.19	<.001	2.65
Embodied rowing	4.92 (.62)	6.22 (.54)	1.31	6.31	<.001	1.82
Cultural exploration	4.54 (.51)	5.79 (.54)	1.25	8.47	<.001	2.45
Artifact experience	4.81 (.67)	6.00 (.70)	1.19	6.28	<.001	1.81
Enjoyment	5.54 (.78)	6.58 (.63)	1.04	7.24	<.001	2.09
Discomfort	1.92 (1.56)	1.75 (1.14)	-.17	-.80	.876	-.23
Time (s)	119.00 (22.38)	119.33 (21.96)	.33	.11	.913	.03

Discomfort remained low in both conditions and did not differ significantly. Completion time also did not differ significantly between conditions. The two conditions used the same rowing control, virtual vessel, route, scene sequence, and completion procedure. Thus, the tangible vessel improved the experiential and cultural dimensions without introducing additional task difficulty, slowing down the route, or increasing discomfort. In this formative test, the physical artifact appeared to change how players interpreted the same rowing journey rather than making the interaction more demanding.

5 Discussion

5.1 A Physical Artifact Can Change the Meaning of the Same VR Content

The tangible and virtual-only conditions used the same virtual route and the same software rowing controller. The main difference was whether the player physically occupied the *Ba He Tong* replica or sat on a regular chair. The significant gains in presence, embodied rowing, cultural exploration, and artifact experience indicate that the tangible vessel changed how participants interpreted the same game content. The physical interface helped the vessel feel less like a generic vehicle and more like a meaningful cultural object.

At the same time, completion time and discomfort did not significantly differ between conditions. This suggests that the tangible setup did not improve the experience by making the game easier, faster, or more mechanically different. Instead, its contribution was experiential and interpretive: it placed the player's body inside the cultural artifact and made the same rowing action feel more connected to the represented water practice.

5.2 Limitations and Future Work

The current game is a stylized interpretation of Jiaomen River rather than a historically verified reconstruction. The tangible condition also combines several physical cues, including enclosure, material,

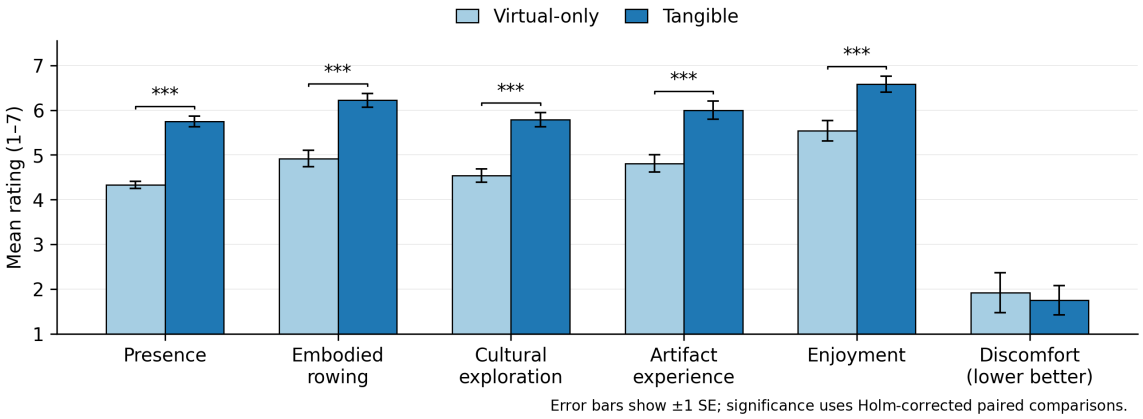


Figure 4: User testing results comparing the virtual-only and tangible *Ba He Tong* setups. Bars show mean ratings, and error bars show ±1 SE. Significance labels indicate Holm-corrected paired comparisons ($^{***}p_H < .001$). Discomfort is marked as lower better.

posture, and artifact identity, so the study cannot isolate which cue contributed most strongly. The sample size is suitable for formative game-design evaluation but not for broad psychological claims. Future work will improve the visual specificity of both scenes, add more environmental storytelling without interrupting the rowing rhythm, and refine the physical installation for easier transport and public exhibition. In addition, future work will explore other gameplay scenarios, such as multi-user play and as part of an exergame [6]. In the context of cultural heritage, it will be interesting to investigate support for intergenerational users [4], which reflects typical visits to museums and cultural sites involving (grand)parent(s) and (grand)child(ren).

Beyond this specific application context, this work suggests a design opportunity for cultural games centered on bodily practices. When a cultural practice involves a recognizable tool, vessel, instrument, or working object, tangible design can make that object part of the player's mode of entry rather than only part of the visual scene. Future cultural game designs could explore similar relationships between physical artifacts, repeated bodily actions, and place-based narratives, while carefully grounding each artifact in its own cultural context.

6 Conclusion

Row Across Time is a tangible VR rowing game that explores how local river culture can be experienced through embodied play. By placing players inside a physical *Ba He Tong* and asking them to row from Old Jiaomen River to Present Jiaomen River, the game turns a cultural artifact into both the subject and interface of play. A formative user study showed that the tangible setup significantly improved presence, embodied rowing, cultural exploration, artifact experience, and enjoyment compared with a virtual-only version, while discomfort and completion time remained comparable. The project demonstrates how cultural heritage games can communicate not only through narrative and scenery, but also through the physical form by which players enter the experience.

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References

[1] Mahdi Azmandian, Mark Hancock, Hrvoje Benko, Eyal Ofek, and Andrew D. Wilson. 2016. Haptic Retargeting: Dynamic Repurposing of Passive Haptics for Enhanced Virtual Reality Experiences. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 1968–1979. doi:10.1145/2858036.2858226

[2] Mafkereseb Kassahun Bekele, Roberto Pierdicca, Emanuele Frontoni, Eva Savina Malinverni, and James Gain. 2018. A Survey of Augmented, Virtual, and Mixed Reality for Cultural Heritage. *ACM Journal on Computing and Cultural Heritage* 11, 2 (2018), 1–36. doi:10.1145/3145534

[3] Erik Champion. 2015. *Critical Gaming: Interactive History and Virtual Heritage*. Routledge.

[4] Hongxi Chen, Jia Zhang, Zixuan Guo, Nilufar Baghaei, and Hai-Ning Liang. 2026. XR Intergenerational Exergaming: Exploring the Complementary Roles of VR, MR, and Environmental Context. *IEEE Transactions on Visualization and Computer Graphics* 32, 5 (2026), 4123–4132. doi:10.1109/TVCG.2026.3680711

[5] Paul Dourish. 2001. *Where the Action Is: The Foundations of Embodied Interaction*. MIT Press.

[6] Wei He, Xiang Li, Mark Colley, Per Ola Kristensson, Ge Lin Kan, and Hai-Ning Liang. 2026. An Exploration of Collaboration Strategies Across Levels of Co-Presence in VR Exergames. *Virtual Reality* (2026).

[7] Eva Hornecker and Jacob Buur. 2006. Getting a Grip on Tangible Interaction: A Framework on Physical Space and Social Interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 437–446. doi:10.1145/1124772.1124838

[8] Brent Edward Insko. 2001. *Passive Haptics Significantly Enhances Virtual Environments*. Ph. D. Dissertation. University of North Carolina at Chapel Hill.

[9] Hiroshi Ishii and Brygg Ullmer. 1997. Tangible Bits: Towards Seamless Interfaces Between People, Bits and Atoms. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*. 234–241. doi:10.1145/258549.258715

[10] Konstantina Kilteni, Raphaela Groten, and Mel Slater. 2012. The Sense of Embodiment in Virtual Reality. *Presence: Teleoperators and Virtual Environments* 21, 4 (2012), 373–387. doi:10.1162/PRES_a_00124

[11] Jiachen Liang, Yue Li, Xueqi Wang, Yuexin Yao, Richard Koeck, and Hai-Ning Liang. 2025. ArtifactVM: Exploring Culturally Meaningful Presentations and User Interactions in Virtual Museums. In *Digital Heritage*, Stefano Campana, Daniele Ferdani, Holger Graf, Gabriele Guidi, Zackary Hegarty, Sofia Pescarin, and Fabio Remondino (Eds.). The Eurographics Association. doi:10.2312/dh.20253267

[12] Yilin Liu, Yiming Lin, Rongkai Shi, Yiming Luo, and Hai-Ning Liang. 2021. RelicVR: A Virtual Reality Game for Active Exploration of Archaeological Relics. In

Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play (Virtual Event, Austria) *(CHI PLAY '21)*. Association for Computing Machinery, New York, NY, USA, 326–332. doi:10.1145/3450337.3483507

[13] Michela Mortara, Chiara Eva Catalano, Francesco Bellotti, Giusy Fiucci, Minica Houry-Panchetti, and Panagiotis Petridis. 2014. Learning Cultural Heritage by Serious Games. *Journal of Cultural Heritage* 15, 3 (2014), 318–325. doi:10.1016/j.culher.2013.04.004

[14] Maria Roussou. 2004. Learning by Doing and Learning Through Play: An Exploration of Interactivity in Virtual Environments for Children. *Computers in Entertainment* 2, 1 (2004), 10. doi:10.1145/973801.973818

[15] Thomas Schubert, Frank Friedmann, and Holger Regenbrecht. 2001. The Experience of Presence: Factor Analytic Insights. *Presence: Teleoperators and Virtual Environments* 10, 3 (2001), 266–281. doi:10.1162/105474601300343603

[16] Adalberto L. Simeone, Eduardo Velloso, and Hans Gellersen. 2015. Substitutional Reality: Using the Physical Environment to Design Virtual Reality Experiences. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 3307–3316. doi:10.1145/2702123.2702389

[17] Mel Slater. 2009. Place Illusion and Plausibility Can Lead to Realistic Behaviour in Immersive Virtual Environments. *Philosophical Transactions of the Royal Society B: Biological Sciences* 364, 1535 (2009), 3549–3557. doi:10.1098/rstb.2009.0138

[18] Hanbing Wang, Junyan Du, Yue Li, Lie Zhang, and Xiang Li. 2025. Grand Challenges in Immersive Technologies for Cultural Heritage. *International Journal of Human–Computer Interaction* 41, 21 (2025), 13682–13703. doi:10.1080/10447318.2025.2475996

[19] Xian Wang, Diego Monteiro, Lik-Hang Lee, Pan Hui, and Hai-Ning Liang. 2022. VibroWeight: Simulating Weight and Center of Gravity Changes of Objects in Virtual Reality for Enhanced Realism. In *2022 IEEE Haptics Symposium (HAPTICS)*. 1–7. doi:10.1109/HAPTICS52432.2022.9765609

[20] Fangtao Zhao, Ziming Li, Yiming Luo, Yue Li, and Hai-Ning Liang. 2025. AirWhisper: enhancing virtual reality experience via visual-airflow multimodal feedback. *Journal on Multimodal User Interfaces* 19, 2 (2025), 139–154. doi:10.1007/s12193-024-00438-9